

Unit 7: First-Order Exactness and Methods

Prepared by Staples Education

Practice Set: Problems and Complete Solutions

Part II — Review Guide: Practice Problems

Work the following ten problems in order. Problems 1–4 develop the core exactness test and potential-function reconstruction (Modules 7.1–7.3). Problems 5–7 require finding a special integrating factor $\mu(x)$ or $\mu(y)$ to repair a non-exact equation (Modules 7.4–7.5). Problem 8 applies the conservative-field interpretation and the Fundamental Theorem for Line Integrals (Module 7.6). Problems 9–10 treat thermodynamic state functions, the exact versus inexact distinction, and entropy via the $1/T$ integrating factor (Module 7.7). Complete solutions follow in Part III.

1. For the equation $(2x + 3y) dx + (3x + 2y) dy = 0$, compute M_y and N_x , confirm the equation is exact, and reconstruct the potential function $F(x, y) = C$.
2. Verify that $(2xy - \sec^2 x) dx + (x^2 + 2y) dy = 0$ is exact, then determine $F(x, y)$ by integrating M with respect to x and solving for the integration term $g(y)$.
3. Show that $(e^x \sin y + 2y) dx + (e^x \cos y + 2x - 2) dy = 0$ is exact and find the implicit solution $F(x, y) = C$.
4. Given $(\frac{y}{x} + 6x) dx + (\ln x - 2) dy = 0$ for $x > 0$, test for exactness, reconstruct $F(x, y)$ including the full evaluation of $g(y)$, and then solve the initial-value problem with $y(1) = 4$.
5. The equation $(3xy + y^2) dx + (x^2 + xy) dy = 0$ is not exact. Determine whether an integrating factor of the form $\mu(x)$ exists, find it, and solve the resulting exact equation.
6. Show that $(2y^2 + 3x) dx + 2xy dy = 0$ fails the exactness test, find an integrating factor depending on x alone, and produce the solution $F(x, y) = C$.
7. For $y dx + (2x - ye^y) dy = 0$, demonstrate that no integrating factor $\mu(x)$ works but an integrating factor $\mu(y)$ does; determine $\mu(y)$ and solve.
8. Consider the vector field $\mathbf{F}(x, y) = \langle 2xy + y^3, x^2 + 3xy^2 \rangle$. Show that $\mathbf{F}$ is conservative, find a scalar potential φ with $\nabla\varphi = \mathbf{F}$, and use it to evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ along any path from $(0, 0)$ to $(2, 1)$.
9. For an ideal gas the inexact heat differential is $\delta Q = C_V dT + \frac{nRT}{V} dV$. Show that δQ is not exact, then verify that multiplying by the integrating factor $1/T$ produces an exact differential $dS = \delta Q/T$, and reconstruct the state function $S(T, V)$.
10. **(Capstone)** Given $dU = C_V dT$ (internal energy) and $\delta Q = dU + P dV$ with $P = nRT/V$ for an ideal gas, determine which of dU and δQ is an exact (state) differential by the $M_y = N_x$ criterion applied to each. Confirm that $1/T$ converts the inexact form into the exact entropy differential dS , and exhibit the recovered $S(T, V)$ up to an additive constant.

Part III — Complete Solutions

Solution 1. Identify $M = 2x + 3y$ and $N = 3x + 2y$. Compute the cross-partial derivatives:

$$M_y = \frac{\partial}{\partial y}(2x + 3y) = 3, \quad N_x = \frac{\partial}{\partial x}(3x + 2y) = 3.$$

Since $M_y = N_x = 3$, the equation is **exact**. Integrate M with respect to x , holding y fixed:

$$F(x, y) = \int (2x + 3y) dx = x^2 + 3xy + g(y).$$

Differentiate F with respect to y and match to N :

$$F_y = 3x + g'(y) = N = 3x + 2y \implies g'(y) = 2y.$$

Integrate $g'(y)$:

$$g(y) = \int 2y dy = y^2.$$

The general solution is

$$F(x, y) = x^2 + 3xy + y^2 = C.$$

Solution 2. Identify $M = 2xy - \sec^2 x$ and $N = x^2 + 2y$. Compute the cross-partial derivatives:

$$M_y = \frac{\partial}{\partial y}(2xy - \sec^2 x) = 2x, \quad N_x = \frac{\partial}{\partial x}(x^2 + 2y) = 2x.$$

Since $M_y = N_x = 2x$, the equation is **exact**. Integrate M with respect to x , using $\int \sec^2 x dx = \tan x$:

$$F(x, y) = \int (2xy - \sec^2 x) dx = x^2 y - \tan x + g(y).$$

Differentiate F with respect to y and match to N :

$$F_y = x^2 + g'(y) = N = x^2 + 2y \implies g'(y) = 2y.$$

Integrate $g'(y)$:

$$g(y) = y^2.$$

The general solution is

$$F(x, y) = x^2 y - \tan x + y^2 = C.$$

Solution 3. Identify $M = e^x \sin y + 2y$ and $N = e^x \cos y + 2x - 2$. Compute the cross-partial derivatives:

$$M_y = \frac{\partial}{\partial y}(e^x \sin y + 2y) = e^x \cos y + 2.$$

$$N_x = \frac{\partial}{\partial x}(e^x \cos y + 2x - 2) = e^x \cos y + 2.$$

Since $M_y = N_x = e^x \cos y + 2$, the equation is **exact**. Integrate M with respect to x , using $\int e^x \sin y dx = e^x \sin y$ (since $\sin y$ is constant in x):

$$F(x, y) = \int (e^x \sin y + 2y) dx = e^x \sin y + 2xy + g(y).$$

Differentiate F with respect to y and match to N :

$$F_y = e^x \cos y + 2x + g'(y) = e^x \cos y + 2x - 2 \implies g'(y) = -2.$$

Integrate $g'(y)$:

$$g(y) = -2y.$$

The general solution is

$$F(x, y) = e^x \sin y + 2xy - 2y = C.$$

Solution 4. Identify $M = y/x + 6x$ and $N = \ln x - 2$ (with $x > 0$). Compute the cross-partial derivatives:

$$M_y = \frac{\partial}{\partial y} \left(\frac{y}{x} + 6x \right) = \frac{1}{x}, \quad N_x = \frac{\partial}{\partial x} (\ln x - 2) = \frac{1}{x}.$$

Since $M_y = N_x = 1/x$, the equation is **exact**. Integrate M with respect to x , using $\int y/x \, dx = y \ln x$ and $\int 6x \, dx = 3x^2$:

$$F(x, y) = \int \left(\frac{y}{x} + 6x \right) dx = y \ln x + 3x^2 + g(y).$$

Differentiate F with respect to y and match to N :

$$F_y = \ln x + g'(y) = \ln x - 2 \implies g'(y) = -2.$$

Integrate $g'(y)$:

$$g(y) = -2y.$$

The general solution is $y \ln x + 3x^2 - 2y = C$. Apply the initial condition $y(1) = 4$; since $\ln 1 = 0$:

$$(4) \ln 1 + 3(1)^2 - 2(4) = 0 + 3 - 8 = -5 = C.$$

The particular implicit solution is $y \ln x + 3x^2 - 2y = -5$. Collecting y terms and solving explicitly:

$$y(\ln x - 2) = -5 - 3x^2 \implies y = \frac{-5 - 3x^2}{\ln x - 2} = \frac{3x^2 + 5}{2 - \ln x}.$$

Solution 5. Identify $M = 3xy + y^2$ and $N = x^2 + xy$. Compute the cross-partial derivatives:

$$M_y = \frac{\partial}{\partial y} (3xy + y^2) = 3x + 2y, \quad N_x = \frac{\partial}{\partial x} (x^2 + xy) = 2x + y.$$

Since $M_y = 3x + 2y \neq 2x + y = N_x$, the equation is **not exact**. Compute the exactness defect:

$$\Delta = M_y - N_x = (3x + 2y) - (2x + y) = x + y.$$

Test for $\mu(x)$ by forming Δ/N :

$$\frac{\Delta}{N} = \frac{x + y}{x^2 + xy} = \frac{x + y}{x(x + y)} = \frac{1}{x}.$$

This is a function of x alone, so $\mu(x)$ exists:

$$\mu(x) = e^{\int (1/x) \, dx} = e^{\ln x} = x.$$

Multiply the ODE by $\mu = x$; the new coefficients are $M^* = x(3xy + y^2) = 3x^2y + xy^2$ and $N^* = x(x^2 + xy) = x^3 + x^2y$. Re-verify exactness:

$$(M^*)_y = \frac{\partial}{\partial y}(3x^2y + xy^2) = 3x^2 + 2xy, \quad (N^*)_x = \frac{\partial}{\partial x}(x^3 + x^2y) = 3x^2 + 2xy.$$

Equal. Now exact. Integrate M^* with respect to x :

$$F(x, y) = \int (3x^2y + xy^2) dx = x^3y + \frac{1}{2}x^2y^2 + g(y).$$

Differentiate F with respect to y and match to N^* :

$$F_y = x^3 + x^2y + g'(y) = x^3 + x^2y \implies g'(y) = 0, \quad g(y) = 0.$$

The general solution is

$$F(x, y) = x^3y + \frac{1}{2}x^2y^2 = C.$$

Solution 6. Identify $M = 2y^2 + 3x$ and $N = 2xy$. Compute the cross-partial derivatives:

$$M_y = \frac{\partial}{\partial y}(2y^2 + 3x) = 4y, \quad N_x = \frac{\partial}{\partial x}(2xy) = 2y.$$

Since $M_y = 4y \neq 2y = N_x$, the equation is **not exact**. Compute the exactness defect:

$$\Delta = M_y - N_x = 4y - 2y = 2y.$$

Test for $\mu(x)$ by forming Δ/N :

$$\frac{\Delta}{N} = \frac{2y}{2xy} = \frac{1}{x}.$$

Pure in x . The integrating factor is:

$$\mu(x) = e^{\int (1/x) dx} = e^{\ln x} = x.$$

Multiply the ODE by $\mu = x$; the new coefficients are $M^* = x(2y^2 + 3x) = 2xy^2 + 3x^2$ and $N^* = x \cdot 2xy = 2x^2y$. Re-verify exactness:

$$(M^*)_y = \frac{\partial}{\partial y}(2xy^2 + 3x^2) = 4xy, \quad (N^*)_x = \frac{\partial}{\partial x}(2x^2y) = 4xy.$$

Equal. Now exact. Integrate M^* with respect to x :

$$F(x, y) = \int (2xy^2 + 3x^2) dx = x^2y^2 + x^3 + g(y).$$

Differentiate F with respect to y and match to N^* :

$$F_y = 2x^2y + g'(y) = 2x^2y \implies g'(y) = 0, \quad g(y) = 0.$$

The general solution is

$$F(x, y) = x^2y^2 + x^3 = C.$$

Solution 7. Identify $M = y$ and $N = 2x - ye^y$. Compute the cross-partial derivatives:

$$M_y = \frac{\partial}{\partial y}(y) = 1, \quad N_x = \frac{\partial}{\partial x}(2x - ye^y) = 2.$$

Since $M_y = 1 \neq 2 = N_x$, the equation is **not exact**. Compute $\Delta = M_y - N_x = 1 - 2 = -1$.

Test $\mu(x)$: Form Δ/N :

$$\frac{\Delta}{N} = \frac{-1}{2x - ye^y}.$$

This contains y , so it is not a function of x alone. **No $\mu(x)$ exists.**

Test $\mu(y)$: Form $-\Delta/M = (N_x - M_y)/M$:

$$\frac{N_x - M_y}{M} = \frac{2 - 1}{y} = \frac{1}{y}.$$

Pure in y . The integrating factor is:

$$\mu(y) = e^{\int (1/y) dy} = e^{\ln y} = y.$$

Multiply the ODE by $\mu = y$; the new coefficients are $M^* = y \cdot y = y^2$ and $N^* = y(2x - ye^y) = 2xy - y^2e^y$. Re-verify exactness:

$$(M^*)_y = \frac{\partial}{\partial y}(y^2) = 2y, \quad (N^*)_x = \frac{\partial}{\partial x}(2xy - y^2e^y) = 2y.$$

Equal. Now exact. Integrate M^* with respect to x :

$$F(x, y) = \int y^2 dx = xy^2 + g(y).$$

Differentiate F with respect to y and match to N^* :

$$F_y = 2xy + g'(y) = 2xy - y^2e^y \implies g'(y) = -y^2e^y.$$

Evaluate $g(y) = -\int y^2e^y dy$ by integration by parts twice. First application, $u = y^2$ and $dv = e^y dy$:

$$\int y^2e^y dy = y^2e^y - 2 \int ye^y dy.$$

Second application for $\int ye^y dy$, with $u = y$ and $dv = e^y dy$:

$$\int ye^y dy = ye^y - \int e^y dy = ye^y - e^y = e^y(y - 1).$$

Combining:

$$\int y^2e^y dy = y^2e^y - 2e^y(y - 1) = e^y(y^2 - 2y + 2).$$

Therefore:

$$g(y) = -e^y(y^2 - 2y + 2).$$

The general solution is

$$F(x, y) = xy^2 - e^y(y^2 - 2y + 2) = C.$$

Solution 8. Let $P = 2xy + y^3$ and $Q = x^2 + 3xy^2$. Compute the cross-partial derivatives:

$$P_y = \frac{\partial}{\partial y}(2xy + y^3) = 2x + 3y^2, \quad Q_x = \frac{\partial}{\partial x}(x^2 + 3xy^2) = 2x + 3y^2.$$

Since $P_y = Q_x = 2x + 3y^2$ on all of $\mathbb{R}^2$ (a simply connected domain), $\mathbf{F}$ is **conservative**. Find the scalar potential φ by integrating P with respect to x :

$$\varphi(x, y) = \int (2xy + y^3) dx = x^2y + xy^3 + g(y).$$

Differentiate φ with respect to y and match to Q :

$$\varphi_y = x^2 + 3xy^2 + g'(y) = x^2 + 3xy^2 \implies g'(y) = 0, \quad g(y) = 0.$$

Therefore:

$$\varphi(x, y) = x^2y + xy^3.$$

By the Fundamental Theorem for Line Integrals, the integral is path independent:

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \varphi(2, 1) - \varphi(0, 0).$$

Evaluate each endpoint:

$$\varphi(2, 1) = (2)^2(1) + (2)(1)^3 = 4 + 2 = 6, \quad \varphi(0, 0) = 0.$$

Therefore:

$$\int_C \mathbf{F} \cdot d\mathbf{r} = 6 - 0 = 6.$$

Solution 9. Treat T and V as the two independent variables. The differential is $\delta Q = C_V dT + \frac{nRT}{V} dV$, so $M = C_V$ (coefficient of dT) and $N = nRT/V$ (coefficient of dV).

Compute the cross-partial derivatives:

$$\frac{\partial M}{\partial V} = \frac{\partial C_V}{\partial V} = 0, \quad \frac{\partial N}{\partial T} = \frac{\partial}{\partial T} \left(\frac{nRT}{V} \right) = \frac{nR}{V}.$$

Since $0 \neq nR/V$, the cross partials are unequal: δQ is **not exact** and heat is not a state function.

Multiply by the integrating factor $1/T$ to define $dS = \delta Q/T$:

$$dS = \frac{\delta Q}{T} = \frac{C_V}{T} dT + \frac{nR}{V} dV.$$

Now $M^* = C_V/T$ and $N^* = nR/V$. Re-verify exactness:

$$\frac{\partial M^*}{\partial V} = \frac{\partial}{\partial V} \left(\frac{C_V}{T} \right) = 0, \quad \frac{\partial N^*}{\partial T} = \frac{\partial}{\partial T} \left(\frac{nR}{V} \right) = 0.$$

Both cross partials equal zero; dS is **exact**, confirming $1/T$ is a valid integrating factor and S is a state function.

Reconstruct S by integrating $M^* = C_V/T$ with respect to T (C_V constant):

$$S(T, V) = \int \frac{C_V}{T} dT + g(V) = C_V \ln T + g(V).$$

Differentiate S with respect to V and match to N^* :

$$\frac{\partial S}{\partial V} = g'(V) = \frac{nR}{V} \implies g(V) = nR \ln V.$$

The entropy state function is:

$$S(T, V) = C_V \ln T + nR \ln V + \text{const.}$$

Solution 10 (Capstone). The internal energy of an ideal gas depends only on temperature (Joule's theorem), so the genuine state-function differential is $dU = C_V dT + 0 \cdot dV$. With $M = C_V$ (coefficient of dT) and $N = 0$ (coefficient of dV), test exactness:

$$\frac{\partial M}{\partial V} = \frac{\partial C_V}{\partial V} = 0, \quad \frac{\partial N}{\partial T} = \frac{\partial(0)}{\partial T} = 0.$$

The cross partials are equal: $dU = C_V dT$ is **exact**. Integrate to confirm the potential:

$$U(T) = \int C_V dT = C_V T + \text{const.}$$

U is a genuine state function depending only on T .

Now build δQ using $dU = C_V dT$ and $P dV = (nRT/V) dV$:

$$\delta Q = dU + P dV = C_V dT + \frac{nRT}{V} dV.$$

With $M = C_V$ and $N = nRT/V$, test exactness:

$$\frac{\partial M}{\partial V} = 0, \quad \frac{\partial N}{\partial T} = \frac{nR}{V}.$$

Since $0 \neq nR/V$, δQ is **not exact**: heat is not a state function. Of the two, dU is the exact (state) differential and δQ is the inexact one.

Apply the integrating factor $1/T$ to δQ :

$$dS = \frac{\delta Q}{T} = \frac{C_V}{T} dT + \frac{nR}{V} dV.$$

Test exactness of dS with $M^* = C_V/T$ and $N^* = nR/V$:

$$\frac{\partial M^*}{\partial V} = 0, \quad \frac{\partial N^*}{\partial T} = 0.$$

Equal. dS is **exact**. Integrate M^* with respect to T :

$$S(T, V) = \int \frac{C_V}{T} dT + g(V) = C_V \ln T + g(V).$$

Match $\partial S/\partial V$ to N^* :

$$g'(V) = \frac{nR}{V} \implies g(V) = nR \ln V.$$

The entropy state function recovered from $\delta Q/T$ is:

$$S(T, V) = C_V \ln T + nR \ln V + \text{const.}$$

Unit 7 Key Takeaway

Every exact equation $M dx + N dy = 0$ reduces to finding the potential $F(x, y) = C$ via the integration loop $F = \int M dx + g(y)$ with $g'(y)$ pinned by matching $F_y = N$. For non-exact equations, compute the defect $\Delta = M_y - N_x$ and apply $\mu(x) = e^{\int \Delta/N dx}$ when Δ/N is pure in x , or $\mu(y) = e^{-\int \Delta/M dy}$ when $-\Delta/M$ is pure in y ; always re-verify exactness after multiplying. The conservative-field identity $M_y = N_x \iff \mathbf{F} = \nabla\varphi$ makes line integrals path-independent: $\int_C \mathbf{F} \cdot d\mathbf{r} = \varphi(B) - \varphi(A)$. In thermodynamics, dU is exact while δQ is not; the integrating factor $1/T$ converts δQ into the exact entropy differential $dS = C_V \ln T + nR \ln V + \text{const.}$